

Multiple Choice (10 marks)

1. What is one condition necessary to produce diamonds?
 - (a) high altitude
 - (b) intense pressure
 - (c) quick cooling
 - (d) low temperatures
2. Heart muscle is best described as
 - (a) the same kind of cells working together.
 - (b) different systems working together.
 - (c) the same kind of organs forming a system.
 - (d) different organs performing a function.
3. Which of the following is NOT a description of compounds?
 - (a) They can exist in the form of atoms or molecules.
 - (b) They consist of atoms of two or more elements bonded together.
 - (c) They have properties that are different from their component elements.
 - (d) They can be broken down into elements by chemical means but not physical means.
4. In a copper wire, a temperature increase is the result of which of the following?
 - (a) an increase in the size of the copper particles
 - (b) a decrease in the mass of the copper particles
 - (c) an increase in the motion of the copper particles
 - (d) a decrease in the distance between the copper particles
5. The gravitational force of the Sun affects the planets in our solar system. Which of these is influenced the most by this force?
 - (a) axial tilt
 - (b) orbital path
 - (c) the masses of the planets
 - (d) the number of moons per planet

True / False (10 marks)

Answer True or False

6. True or False: Hot air warming a room is an example of heat transfer by radiation.
7. True or False: If a nerve cell stops functioning, it will start dividing into more cells.
8. True or False: A nail being hammered demonstrates the use of mechanical energy as it involves physical movement to drive the nail into a surface.
9. True or False: In January, Gemini typically appears in the sky after midnight.
10. True or False: The distance from the Sun to Earth is approximately 5.0×10^2 light-seconds.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the types of evidence that could indicate a bird species, last seen in the 1900s, may still exist. How would this evidence challenge the claim of its extinction?
12. Discuss the ecological consequences of preventing natural forest fires in a climax fire ecosystem and explain how this might affect its stability and biodiversity.

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